

than 200 metres. One RS-Ruby on top of the vehicle has a vertical field-of-view (FOV) from -25° to $+15^{\circ}$, but there is still a small blind spot less than 0.4 metres around the vehicle body. Thus, four RS-BPearl are embedded sideways around the vehicle to cover the small blind zones to guarantee a complete 360-degree surrounding view.

RS-Fusion-P5 fuses point from RS-Ruby and RS-BPearl in real time, and generates laser points of more than 4,600,000 per second to identify all-around obstacles, and position easily and precisely. Through its advanced AI perception algorithms, synchronisation interfaces and multi-sensor fusion, it achieves unprecedented safety and reliability.

RS-Ruby has 128 beams for L4 and L5 autonomous driving with a vertical resolution of 0.1° , which fully meets the further demand of high-speed autonomous driving. It not only meets the automotive-grade requirement of

working from -40°C to 60°C but also achieves breakthrough performance in all-weather conditions. It achieves anti-interference of multiple lidar using special laser encryption technology to filter interference signals. Besides, it is not affected by strong direct sunlight. Lidar reflection intensity reaches a perfect balance of consistency and distinction.

RS-BPearl has fine near-field detection of $360^{\circ} \times 90^{\circ}$ super-wide FOV, and short-range blind spot lidar, which can precisely identify objects around the vehicle body such as pedestrians, pets, vehicles, roadbeds as well as other details of the near-field ground area. It can also detect actual height information under particular scenarios such as bridge tunnels and culverts for improving driving decisions and car safety.

There are also other solutions including RS-P1 mature and reliable lidar perception solution for low-speed autonomous driving, RS-P2

lidar perception solution for medium-high speed autonomous driving and RS-Fusion-P3 full-stack lidar perception solution for L3+ autonomous driving.

Dr Leilei Shinohara, vice president - R&D, RoboSense, says, "Driving safety is the most critical challenge for autonomous driving vehicles. Perception is the first sense for autonomous vehicles to precept the surrounding environment." The cold-resistant lidar maps the 3D world through emitting and receiving laser pulses. With point cloud algorithms it can accurately recognise obstacles, even in snow and ice. Therefore lidar is the most important sensor when it comes to bringing autonomous driving technology to extremely cold environments. Point cloud data quality on rainy days is excellent and outperforms state-of-the-art options to better cope with harsh environments and extreme conditions.

—Deepshikha Shukla

4 SWEEPING AND MOPPING ROBOT TO FREE PEOPLE FROM TRIVIAL HOUSEWORK

You no longer have to dirty your hands while mopping. The pain point of traditional mopping is the dirty cloth that becomes dirtier during usage. The dirty cloth needs to be handwashed, which not only dirties hands but also takes time and effort. Venii N1bot returns for cleaning after every 20m^2 to mop the whole room with clean cloth. Its self-cleaning makes mopping easy and worry-free. The bot is a fully-automatic mopping robot system.

Venii N1bot promotes artificial intelligence (AI) and electromechanical technology to realise the benefits of home automation and free people from trivial housework. Besides the innovative self-washing technology, it leads in the fields of embedded machine vision and AI by leveraging multiple core invention patents.

N1 can mop more effectively by applying 10N pressure, generated from its Lift-Mop patented mop-



ping mechanical structure—meaning motor torque of the crank connecting rod and spring inside the pressure rod—which makes it different from normal mopping.

To wash the fabric with clean water, clean and dirty water is separated within the wash base. N1's 800-millilitre double watertank enables 120m^2 with one full tank.

A traditional cleaner collects dust into a dust box, which is hard to clean and may dirty hands. N1 is equipped

with a large disposable dust bag of about one litre capacity. Dirty water can be discarded easily when the tank is full, without getting the hands dirty.

N1 gives stronger sweeping performance with SLAM mapping and corner cleaning with a retractable side brush. Its advanced lidar SLAM technology helps clean the whole room with precise navigation and path planning. The bot can clean in zig-zag manner or by dividing spaces